GAME THEORY & NASH EQUILIBRIA — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: April 17, 2026

Diagram-based questions on Game Theory & Nash Equilibria

Question 1

10.0 points

Consider the extensive form game tree for a sequential two-player game shown in the diagram below. Player 1 moves first, choosing between A and B. If Player 1 chooses A, Player 2 chooses between C and D; if B, Player 2 chooses between E and F. Analyze the game tree and identify the Nash equilibrium(s). Assume the following payoffs: $(3, 2)$, $(1, 1)$, $(2, 3)$, and $(0, 0)$.

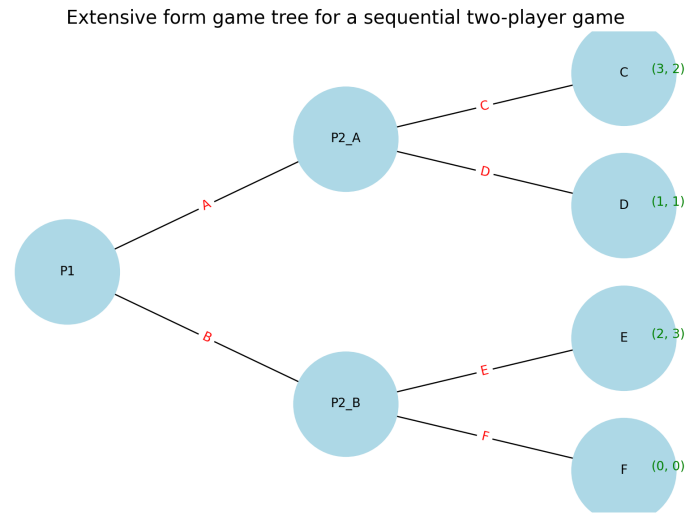

Fig. 1. Extensive form game tree for a sequential two-player game with specified payoffs. Player 1 chooses between A and B; Player 2 chooses between C, D, E, and F based on Player 1's choice. Payoffs are (3, 2), (1, 1), (2, 3), and (0, 0). FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • nash equilibrium The diagram will FAIL review if any of these appear.

Question 2

10.0 points

Analyze the payoff matrix shown in the diagram below for the classic Prisoner's Dilemma. Identify all Nash equilibria for the matrix: Player 1 choices (Cooperate, Defect) and Player 2 choices (Cooperate, Defect) with associated payoffs as follows: $(3, 3)$, $(0, 5)$, $(5, 0)$, and $(1, 1)$.

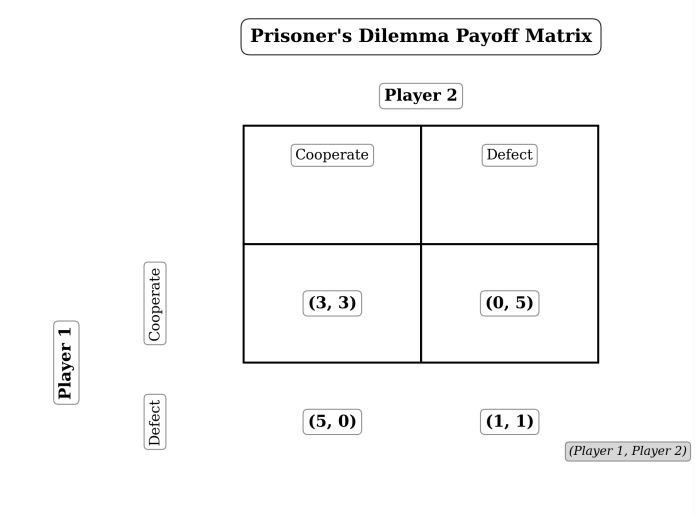

Fig. 2. Payoff matrix for the Prisoner's Dilemma game. Player 1 choices are 'Cooperate' and 'Defect', Player 2 choices are 'Cooperate' and 'Defect'. The payoffs are (3, 3) for both cooperating, (0, 5) for Player 1 cooperating and Player 2 defecting, (5, 0) for Player 1 defecting and Player 2 cooperating, and (1, 1) for both defecting. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • nash equilibrium The diagram will FAIL review if any of these appear.

Question 3

10.0 points

Given a coordination game with the following mixed strategy Nash equilibrium: Two players each choose between actions X and Y. The payoff matrix is defined as shown in the diagram below: $(2, 2)$, $(1, 1)$, $(1, 1)$, and $(3, 3)$. Calculate the probabilities of each player choosing X and Y in the mixed strategy Nash equilibrium.

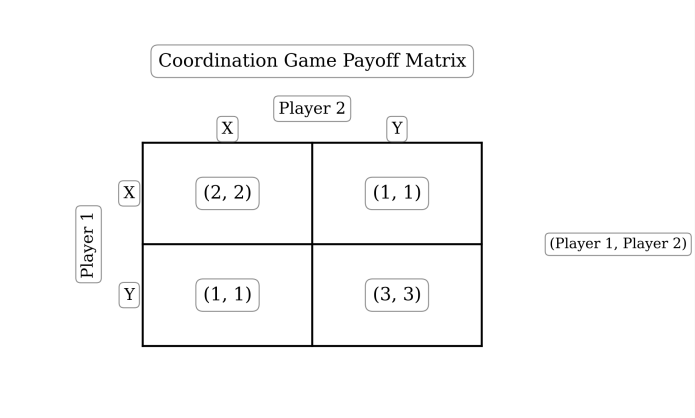

Fig. 3. Payoff matrix for a coordination game with two players. Each player chooses between actions X and Y. The matrix is 2x2 with the following payoffs: (2, 2) for (X, X), (1, 1) for (X, Y) and (Y, X), and (3, 3) for (Y, Y). FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • nash equilibrium The diagram will FAIL review if any of these appear.

Question 4 10.0 points

Using the diagram below of a finite game tree, perform backward induction to determine the optimal strategies for the players. Assume initially that Player 1 can choose between G and H, and Player 2 can choose I or J after observing Player 1's choice. The final payoffs are $(4, 4)$, $(2, 3)$, $(3, 2)$, and $(1, 1)$.

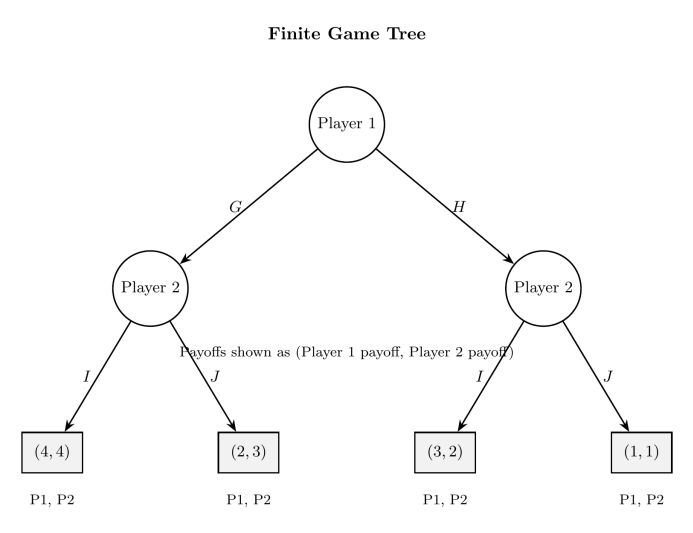

Fig. 4. Finite game tree showing Player 1's choices (G, H) and Player 2's choices (I, J) with final payoffs (4, 4), (2, 3), (3, 2), and (1, 1). FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • optimal The diagram will FAIL review if any of these appear. IMPORTANT: Values or sequences in the diagram don't match the question. Specifically: The diagram shows payoffs (3, 2) and (1, 1) resulting from Player 2's choice 'J. The diagram's structure does not correctly represent the game described in the question, leading to a mismatch between the paths and the specified payoffs.. Regenerate using EXACTLY the values and sequences specified.

Question 5 10.0 points

Examine the Pareto frontier in the diagram below illustrating possible outcomes for two firms. Analyze the diagram to distinguish between efficient and inefficient outcomes. The diagram shows combinations of payoffs $(5, 5)$, $(4, 3)$, and $(2, 4)$ are labeled.

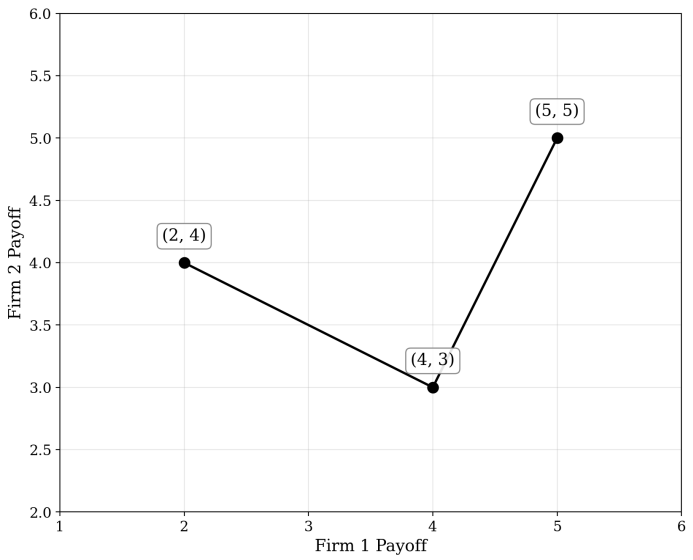

Fig. 5. Pareto frontier diagram for two firms' outcomes, showing combinations of payoffs (5, 5), (4, 3), and (2, 4). The diagram illustrates efficient and inefficient outcomes on a graph with labeled points. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • pareto front • pareto frontier The diagram will FAIL review if any of these appear.